CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech)
University Examination November 2010
CE202 Data Communication and Networking

Date: 27.11.10 , Saturday **Time:10.30 a.m to 1.30 p.m** **Maximum Marks:70**

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1 Answer the following questions. **11**

1. Bits from physical layer are converted into frames by the _______ layer.
2. Network layer of the OSI reference model provides and end-to-end connection. [True/False]
3. HTTP works at the Application layer of the TCP/IP model. [True/False]
4. ______ is a device that transmits or receives data in the form of an analog or digital signal through a network.
5. Which address is used to locate application running on remote machine?
6. Define Baud Rate.
7. __________ is connection oriented protocol at transport layer.
8. _________ is the process in which the receiver tries to guess the message by redundancy bits.
9. FDM is a digital multiplexing technique. [True/False]
10. Define Composite Signals.
11. The period of a signal is 1000 ms. Calculate the frequency in kHz.

Q.2
(a) Explain OSI reference model. Which layers of OSI are not available in TCP/IP model? **8**
(b) Write down the advantages and disadvantages of Mesh and Star topology. **4**

OR

Q.2
(a) What is protocol? Explain key elements of a protocol. What are the two categories of data communication standards? **6**
(b) Define the following addresses in TCP/IP. **6**
1. Physical Address
2. Logical Address
3. Port Address

Q.3
(a) Describes various categories of network like LAN, MAN and WAN with example. **6**
(b) Explain Time Division Multiplexing. **6**

OR

Q.3
(a) What is guided transmission media? Write down the comparison of coaxial cable and twisted-pair. **6**
(b) Why does redundant bits require in Data Link Layer frame header? Compare LRC vs. VRC techniques. **6**

SECTION – II

Q.4 Do as directed. **11**

1. Which type of error can not be detected by checksum?
2. Define Piggybacking.
3. Stop and Wait is a special case of Go Back N with window size 1. [True/False]
4. To create mesh topology with 5 nodes for simplex transmission media, total no of cables required is ___________.
5. What is hamming distance for following codeword? d(10101, 10010)
6. Bit stuffing is the process of adding 1 extra byte whenever there is an escape character in data. [True/False]
7. Bridge works on ___________ layer.
8. Public Switched Telephone Network (PSTN) uses packet switching technique. [True/False]
9. HTTP works on __________ port number.
10. What is an RARP?
11. IP is connection oriented protocol. [True/False]

Q.5

(a) Write down the comparisons of **6**
1. Analog and Digital signals 2. Periodic and Aperiodic signals

(b) What is transmission impairment? Explain the causes of transmission impairment. **6**

OR

Q.5

(a) Explain following terms with example: **6**
1. Amplitude 2. Frequency 3. Phase

(b) Explain Amplitude modulation with necessary diagram. **6**

Q.6 Answer the following questions (Any *TWO*). **12**

(a) Given a 10-bit sequence 1010011110 and a divisor of 1011, find the codeword.

(b) Explain the various classes of Internet Protocol addresses.

(c) Write a short note on Cable Modem.